Fixed-Interior Sharp Minimax Rates for Average Treatment Effects with Unrestricted Discrete Confounding

July 18, 2026

Abstract

This paper studies average treatment effect estimation with binary treatment and outcome when the covariate is a discrete variable with d categories and otherwise unrestricted marginal distribution. Under iid sampling and uniform overlap, the estimand is the covariate-adjusted ATE, written as a sum of homogeneous four-cell functionals. For each fixed overlap constant $0 < \epsilon < 1/2$, there are constants $\rho_\epsilon > 0$ and $N_\epsilon < \infty$ such that, for $n \geq N_\epsilon$, $d > 0$, and $d \leq \rho_\epsilon n \log n$, we establish the minimax mean-squared-error rate

$$\mathsf{R}_{n,d,\epsilon} \asymp_\epsilon \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

Under the same fixed-interior calibration conditions, the upper bound is attained by a computable split-sample hybrid estimator: empirical ratios are used on pilot-certified heavy categories, while light categories are estimated by a Chebyshev-polynomial continuation implemented through factorial moments. The lower-bound scale is imported from the fixed-sample construction of Zeng et al. [2024]; the matching computable upper bound, the fixed-interior rate statement, and the resulting parametric-rate and consistency regimes are proved in this paper. The result implies a parametric-rate window $d = O(\sqrt{n} \log n)$, within the fixed-interior range, and a consistency threshold $d = o(n \log n)$. For each fixed interior $0 < \epsilon < 1/2$, under the same hybrid calibration range $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, we also record an oracle-calibrated deterministic comparison envelope near exact randomization. This theoretical comparison uses the known constants ϵ and C_ϵ. At the separate endpoint $\epsilon = 1/2$, the minimax risk is bracketed between constants times n^{-1}, uniformly in d.

1 Introduction

Covariate adjustment is central to observational causal inference. Under consistency, conditional exchangeability, and overlap, the average treatment effect can be identified by comparing treated and control outcomes within covariate strata and averaging over the covariate distribution [Rubin, 1974, Rosenbaum and Rubin, 1983, Hahn, 1998, Hirano et al., 2003]. Classical semiparametric theory describes this problem well when the adjustment dimension is fixed or when nuisance functions can be estimated with enough structure. This paper studies a finite-alphabet benchmark in which that structure is deliberately absent.

The model is simple. A single observation is $O = (X, A, Y)$, where $X \in \{1, \dots, d\}$, $A \in \{0, 1\}$, and $Y \in \{0, 1\}$. The observations are iid as in assumption 1. The covariate marginal is unrestricted, and treatment assignment satisfies the categorywise overlap condition in assumption 2. With the usual potential-outcome interpretation supplied by assumption 3 and assumption 4, the

ATE is the functional in definition 6. The statistical object is the minimax mean-squared-error risk over the finite-alphabet overlap class in definition 3 and definition 14.

The main result is that, for each fixed $0 < \epsilon < 1/2$, the sharp minimax rate is $n^{-1} + d^2/(n^2 \log^2 n)$ under the stated fixed-interior calibration range. Theorem 1 shows that there are constants depending on ϵ such that, whenever n is large enough, $d > 0$, and $d \leq \rho_\epsilon n \log n$,

$$\mathsf{R}_{n,d,\epsilon} \asymp_\epsilon \frac{1}{n} + \frac{d^2}{n^2(\log n)^2}.$$

Within the same fixed-interior range, the theorem also gives a parametric-rate window when $d \leq K\sqrt{n}\log n$, and it gives the consistency threshold $d = o(n \log n)$. These statements are uniform over unrestricted covariate marginals satisfying only the stated overlap restriction.

The upper bound is constructive. Definition 11 gives a split-sample estimator that first uses pilot counts to classify covariate categories as heavy or light. Heavy categories are estimated by empirical ratios on an independent split. Light categories are estimated by a polynomial approximation to the homogeneous cell functional in definition 5, with factorial moments used to estimate the polynomial terms without unstable empirical denominators. Theorem 2 records both the fixed-overlap risk envelope and the real-arithmetic computability bound for this estimator.

The estimator uses polynomial approximation on weakly sampled cells, a device related to large-alphabet nonsmooth functional estimation [Paninski, 2003, Jiao et al., 2015, Wu and Yang, 2016, Han et al., 2020]. Here the cell contribution is the four-cell ATE term

$$p_k(\mu_{1k} - \mu_{0k}).$$

Zeng et al. [2024] provide the fixed-sample lower-bound ingredient used here and analyze standard regression, weighting, and doubly robust estimators for this high-dimensional discrete ATE problem. This paper constructs the ratio-polynomial hybrid estimator and derives the fixed-interior consistency and parametric-rate regimes. The lower-bound transfer itself is stated in lemma 4.

The result provides a complementary finite-alphabet minimax benchmark for semiparametric and doubly robust theory. In settings with smoothness, sparsity, or accurate nuisance regression and propensity estimates, inverse-probability, doubly robust, targeted, double-machine-learning, and higher-order approaches address different regimes [Robins et al., 1994, Bang and Robins, 2005, van der Laan and Rubin, 2006, Belloni et al., 2014, Chernozhukov et al., 2018, Robins et al., 2008, 2009, 2017, Kennedy et al., 2024]. In the growing unrestricted-alphabet regime studied here, overlap is the sole regularity condition and many small categories create the nonregular term $d^2/(n^2 \log^2 n)$.

The endpoint $\epsilon = 1/2$ is different. At exact within-category randomization, theorem 2 brackets the minimax risk between $1/(100n)$ and $1/n$, uniformly in d. For each fixed interior $0 < \epsilon < 1/2$, and only under the same calibration conditions $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, the same theorem also gives an oracle-calibrated deterministic comparison envelope near randomization,

$$\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2\right\},$$

up to constants. This comparison chooses between two already analyzed estimators using the known fixed-interior constants and supplies an oracle-calibrated upper envelope near the randomized endpoint.

The formal statements and derivations are supported by a Lean 4 development, with the precise verification boundary recorded in the verification appendix.

2 Setup and assumptions

We work with a finite-alphabet observational model. A single observation is $O_i = (X, A, Y)$, where $X \in \{1, \ldots, d\}$ is a discrete covariate, $A \in \{0, 1\}$ is treatment, and $Y \in \{0, 1\}$ is the observed outcome. The finite-alphabet i.i.d. observational world is deliberately nonparametric in the distribution of X: the cell probabilities may be arbitrary subject only to normalization and the overlap restriction introduced below.

Definition 1. [`def:finite-alphabet-probability-laws`] (Finite-alphabet probability laws Prob) *We say $P \in \mathsf{Prob}(\{1, \ldots, d\} \times \{0, 1\}^2)$ when P is a probability measure on the finite observation alphabet $\{1, \ldots, d\} \times \{0, 1\} \times \{0, \ldots, 1\}$, equivalently a nonnegative mass assignment to each cell (k, a, y) whose total mass over all $k \in \{1, \ldots, d\}$, $a \in \{0, 1\}$, and $y \in \{0, 1\}$ is one.*

For P as in definition 1, write q_{aky} for $P(X = k, A = a, Y = y)$, p_k for $P(X = k)$, π_k for $P(A = 1 \mid X = k)$ when $p_k > 0$, and μ_{ak} for $E_P[Y \mid A = a, X = k]$ when the conditioning event has positive probability. We also write q_k for the four-vector $(q_{k00}, q_{k01}, q_{k10}, q_{k11})$, ordered by treatment and outcome. The sample size n is fixed throughout the experiment.

Assumption 1. [`ass:iid-sampling`] (Iid sampling) *The observations $O_1, \ldots, O_n$ are independent and identically distributed with single-observation law P.*

assumption 1 is the standard independent and identically distributed sampling condition; it isolates the difficulty of unrestricted discrete confounding from dependence or sampling-design complications [Zeng et al., 2024].

Treatment assignment is summarized category by category.

Definition 2. [`def:bernoulli-law`] (Bernoulli law $\mathsf{Bernoulli}(\pi_k)$) *We say $A \mid X = k \sim \mathsf{Bernoulli}(\pi_k)$ when $A \in \{0, 1\}$ and, conditional on $X = k$, $P(A = 1 \mid X = k) = \pi_k$ and $P(A = 0 \mid X = k) = 1 - \pi_k$.*

Assumption 2. [`ass:overlap`] (Uniform overlap) *For every k with $p_k > 0$,*

$$\epsilon \leq \pi_k \leq 1 - \epsilon.$$

assumption 2 is the standard strong overlap condition, requiring every observed covariate category to retain treatment and control observations with probability bounded away from zero [Zeng et al., 2024]. The constant ϵ may affect constants in the fixed-interior theory, while the estimator defined below uses the same formula across overlap classes.

For causal interpretation, we use the usual potential-outcome overlay only where needed.

Assumption 3. [`ass:consistency`] (Consistency) *The observed outcome satisfies*

$$Y = Y(A)$$

almost surely.

assumption 3 is the standard consistency condition, linking the realized observed outcome to the potential outcome under the assigned treatment [Zeng et al., 2024]. The potential outcomes $Y(a)$ are binary in the present model.

Assumption 4. [`ass:conditional-exchangeability`] (Conditional exchangeability) *The binary potential outcomes satisfy*

$$(Y(0), Y(1)) \perp A \mid X.$$

assumption 4 is the standard conditional exchangeability condition, under which comparisons within a covariate category identify the category-specific causal contrast [Zeng et al., 2024]. Together with assumption 3 and assumption 2, it gives the observational ATE functional its causal interpretation.

The statistical problem is indexed by all finite-alphabet laws obeying these restrictions.

Definition 3. [`def:experiment-class`] (Experiment class $\mathcal{E}_{n,d,\epsilon}$) *Under assumption 1 and assumption 2, the experiment class is*

$$\mathcal{E}_{n,d,\epsilon} := \{P^{\otimes n} : P \in \mathsf{Prob}(\{1,\ldots,d\}\times\{0,1\}^2),\ n,d \in \mathbb{N},\ 0<\epsilon\le 1/2,\ P \textit{ satisfies assumption 2}\}.$$

The class $\mathcal{E}_{n,d,\epsilon}$ in definition 3 allows the covariate marginal to vary freely over the d categories. Thus sparsity of categories is part of the estimation problem rather than a regularity condition imposed away.

It is convenient to express each covariate category through its four observed cell masses.

Definition 4. [`def:overlap-cone`] (Overlap cone $\mathcal{C}_\epsilon$) *The overlap-restricted cone of nonnegative four-cell vectors is*

$$\mathcal{C}_\epsilon := \{u=(u_{00},u_{01},u_{10},u_{11}) \in \mathbb{R}^4_+ : \epsilon(u_{00}+u_{01}+u_{10}+u_{11}) \le u_{10}+u_{11} \le (1-\epsilon)(u_{00}+u_{01}+u_{10}+u_{11})\}.$$

definition 4 records overlap as a constraint on a nonnegative four-cell vector $\mathcal{C}_\epsilon$. The cone formulation keeps the category mass and the treatment split in the same object, which is useful when categories have small or vanishing probability.

Definition 5. [`def:cell-functional`] (Cell functional $\phi(u)$) *For $u=(u_{00},u_{01},u_{10},u_{11}) \in \mathcal{C}_\epsilon$, with $\mathcal{C}_\epsilon$ as in definition 4, define*

$$\phi(u) = (u_{00}+u_{01}+u_{10}+u_{11})\left\{\frac{u_{11}}{u_{10}+u_{11}} - \frac{u_{01}}{u_{00}+u_{01}}\right\}$$

when $u \neq 0$, and set $\phi(0,0,0,0)=0$.

The homogeneous contribution ϕ in definition 5 is the category mass multiplied by the treated-minus-control conditional mean contrast. Its homogeneity is the source of the nonsmooth behavior at small cell masses.

Definition 6. [`def:ate-functional`] (ATE functional $\tau(P)$) *The average treatment effect functional is*

$$\tau(P) = \sum_{k=1}^d p_k(\mu_{1k}-\mu_{0k}) = \sum_{k=1}^d \phi(q_k).$$

The estimand $\tau(P)$ in definition 6 is therefore both the usual covariate-adjusted ATE and a sum of four-cell contributions. This second representation is the one used by the estimator and by the minimax analysis.

The estimator separates categories using a deterministic half-sample split.

Definition 7. [`def:balanced-sample-split`] (Balanced sample split I_j) *For $j \in \{0,1\}$, the balanced deterministic half-sample index set $I_j \subseteq \{1,\ldots,n\}$ is $I_j = \{i \in \{1,\ldots,n\} : i \le \lfloor n/2 \rfloor\}$ when $j=0$, and $I_j = \{i \in \{1,\ldots,n\} : \lfloor n/2 \rfloor < i \le n\}$ when $j=1$.*

Definition 8. [`def:split-cell-counts`] (Split cell counts $N^{(0)}_{aky}$ and $N^{(1)}_{aky}$) *For $j \in \{0,1\}$, $k \in \{1,\ldots,d\}$, and $a, y \in \{0,1\}$, the split cell count $N^{(j)}_{aky}$ is the number of observations in the deterministic split I_j with covariate value k, treatment value a, and outcome value y:*

$$N^{(j)}_{aky} := \sum_{i \in I_j} \mathbf{1}\{X_i = k,\ A_i = a,\ Y_i = y\}.$$

The first split is used only to classify categories as empirically heavy or light, while the second split estimates their contributions. The count notation $N^{(j)}_{aky}$ in definition 8 will also be used to form factorial moments; for a nonnegative integer z and order r, the falling factorial $(z)_r$ is defined next.

Definition 9. [`def:chebyshev-polynomial`] (Chebyshev Polynomial T_M) *We write T_M for the degree-M Chebyshev polynomial of the first kind on $\mathbb{R}$, characterized by $T_0(x) = 1$, $T_1(x) = x$, and $T_{M+2}(x) = 2xT_{M+1}(x) - T_M(x)$ for all $M \geq 0$ and $x \in \mathbb{R}$.*

Definition 10. [`def:falling-factorial`] (Falling factorial $(x)_r$) *For $x, r \in \mathbb{N}_0$, we write $(x)_r$ for the falling factorial of x of order r, defined by $(x)_r = \prod_{\ell=0}^{r-1}(x-\ell)$, with the empty product convention $(x)_0 = 1$; hence $(x)_r = 0$ when $r > x$.*

The Chebyshev polynomial in definition 9 supplies the light-cell approximation, and definition 10 supplies the unbiased factorial-moment representation for its polynomial terms. For the multi-index notation below, $r = (r_{ay})_{a,y\in\{0,1\}}$ ranges over four nonnegative integer coordinates.

Definition 11. [`def:hybrid-estimator-handle`] (Hybrid estimator $\widehat{\tau}^{\mathrm{hyb}}_n$) *For a sample $O_1, \ldots, O_n$ with $O_i = (X_i, A_i, Y_i) \in \{1,\ldots,d\} \times \{0,1\} \times \{0,1\}$, set*

$$I_0 = \{1,\ldots,\lfloor n/2 \rfloor\}, \qquad I_1 = \{\lfloor n/2 \rfloor + 1, \ldots, n\}, \qquad m_j = |I_j|, \qquad L = \log(en).$$

Let $A = 6$, and set

$$\lambda_0 = 256, \qquad b_0 = 4096, \qquad D_A = 8\log(9A/8) = 8\log(27/4),$$

and

$$\alpha_0 = \min\left\{1, \frac{1}{32\log A}, \frac{1}{256 D_A}\right\}.$$

For $n \geq 2$, define

$$M(n) = \max\{2, \lfloor \alpha_0 L \rfloor\}, \qquad B(n) = b_0 L / m_1.$$

Let N_0 be the least numerical cutoff such that, for every $n \geq N_0$,

$$\alpha_0 L \geq 2, \qquad 4M(n)^2 \leq m_1, \qquad 4M(n)/m_1 \leq 3B(n)/4.$$

For $n \geq N_0$, write $M = M(n)$ and $B = B(n)$, and define the pilot-certified heavy and light category sets by

$$\widehat{\mathcal{H}}_n = \left\{k : \sum_{a,y} N^{(0)}_{aky} > \lambda_0 L\right\}, \qquad \widehat{\mathcal{L}}_n = \widehat{\mathcal{H}}_n^c.$$

For the second split, put

$$N^{(1)}_k = \sum_{a,y} N^{(1)}_{aky}, \qquad N^{(1)}_{ak} = \sum_y N^{(1)}_{aky},$$

and define the heavy-category ratio contribution

$$\widehat{h}_k = \frac{N_k^{(1)}}{m_1}\left\{\frac{N_{1k1}^{(1)}}{N_{1k}^{(1)}}\mathbf{1}\{N_{1k}^{(1)} > 0\} - \frac{N_{0k1}^{(1)}}{N_{0k}^{(1)}}\mathbf{1}\{N_{0k}^{(1)} > 0\}\right\}.$$

Let

$$H_M(x) = \frac{1 - T_M(1-2x)}{2M^2}, \qquad E_M(x) = \frac{H_M(x)}{x}, \qquad G_M(x) = \frac{1 - E_M(x)}{x},$$

where E_M *and* G_M *are interpreted by polynomial continuation at zero. For* $u = (u_{00}, u_{01}, u_{10}, u_{11})$*, set*

$$s_a(u) = u_{a0} + u_{a1}, \qquad s(u) = s_0(u) + s_1(u), \qquad t_1(u) = u_{11}, \qquad t_0(u) = u_{01}.$$

Define the light-cell polynomial approximation

$$P_{M,B}(u) = \frac{s(u)t_1(u)}{B} G_M\left(\frac{s_1(u)}{B}\right) - \frac{s(u)t_0(u)}{B} G_M\left(\frac{s_0(u)}{B}\right).$$

Expanding $P_{M,B}(u) = \sum_{|r| \le M} c_r u^r$*, set*

$$\widehat{P}_k = \sum_{|r| \le M} c_r \frac{\prod_{a,y} (N_{aky}^{(1)})_{r_{ay}}}{(m_1)_{|r|}}, \qquad \widehat{T}_{\mathcal{H}} = \sum_{k \in \widehat{\mathcal{H}}_n} \widehat{h}_k, \qquad \widehat{T}_{\mathcal{L}} = \sum_{k \in \widehat{\mathcal{L}}_n} \widehat{P}_k.$$

The truncated balanced ratio-polynomial hybrid estimator is

$$\widehat{\tau}_n^{\mathrm{hyb}} = \max\{-1, \min(1, \widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}})\}.$$

For $n < N_0$*, set* $\widehat{\mathcal{L}}_n = \varnothing$ *and use the ratio branch on every category. The definition uses no overlap parameter.*

definition 11 combines a ratio estimator on pilot-certified heavy categories with a polynomial estimator on pilot-certified light categories. The sets $\widehat{\mathcal{H}}_n, \widehat{\mathcal{L}}_n$ are determined from I_0, while the light contribution $\widehat{T}_{\mathcal{L}}$ and the final estimator $\widehat{\tau}_n^{\mathrm{hyb}}$ use the second split for estimation. The truncation simply enforces the natural range of an ATE with binary outcomes.

Throughout the reader-facing statistical statements, $\mathbb{N} = \{1, 2, \ldots\}$, so both the sample size and alphabet size are positive. For the operation-count notation set $M(1) = 2$. When $1 \le n < N_0$, the ratio-only branch is evaluated directly; the light-cell quantities $B(n)$, polynomial coefficients, and factorial-moment terms begin at the large-sample branch. The statistical experiment stated here therefore uses positive sample and alphabet sizes, alongside Lean's total conventions at zero.

For comparison near randomized treatment, we also record a centered linear estimator.

Definition 12. [`def:centered-estimator`] (Centered estimator $\widehat{\tau}_{\mathrm{ctr}}$) *For observations* $O_i = (X_i, A_i, Y_i)$*,* $i = 1, \ldots, n$*, with binary treatment* A_i *and binary outcome* Y_i*, the centered estimator is*

$$\widehat{\tau}_{\mathrm{ctr}} = \frac{2}{n}\sum_{i=1}^{n}(2A_i - 1)\left(Y_i - \frac{1}{2}\right).$$

Definition 13. [def:selected-estimator] (Selected estimator $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$) *For fixed* $0 < \epsilon < 1/2$ *and fixed-interior upper-bound constant* C_ϵ*, define* $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$ *to be* $\widehat{\tau}^{\mathrm{hyb}}_n$ *when*

$$C_\epsilon\{1/n + d^2/(n^2\log^2 n)\} \le 1/n + 4(1/2-\epsilon)^2,$$

and to be $\widehat{\tau}_{\mathrm{ctr}}$ *otherwise. At* $n = 1$*, the quotient involving* $\log n$ *is interpreted as zero; the asymptotic results use this selector only above their stated large-sample cutoff.*

This is an oracle-calibrated deterministic comparison rule. It combines the fixed-interior hybrid bound with the near-randomization linear bound using the fixed overlap constant and the fixed-interior upper-bound constant C_ϵ.

The risk criterion is worst-case mean squared error over the overlap experiment class.

Definition 14. [def:minimax-risk] (Minimax risk $\mathsf{R}_{n,d,\epsilon}$) *The minimax mean-squared-error risk is*

$$\mathsf{R}_{n,d,\epsilon} = \inf_{\widehat{\tau}} \sup_{P^{\otimes n}\in\mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}-\tau(P))^2\right],$$

where the infimum ranges over all measurable estimators $\widehat{\tau}$ *based on* $O_1,\ldots,O_n$*.*

The quantity $\mathsf{R}_{n,d,\epsilon}$ in definition 14 is the object characterized in the main results. The supremum is over product laws in definition 3, so the risk treats the covariate distribution and all conditional outcome probabilities as unknown nuisance features.

We close the setup with a two-category construction showing why covariate adjustment is essential even in the smallest nontrivial alphabet.

Definition 15. [def:two-category-witness] (Two-category witness law) *For* $d = 2$*, set*

$$p_1 = p_2 = \frac{1}{2}, \qquad \pi_1 = \epsilon, \qquad \pi_2 = 1-\epsilon, \qquad \mu_{01} = \mu_{02} = 0, \qquad \mu_{11} = 1, \qquad \mu_{12} = 0.$$

Define a full-data extension by drawing

$$(Y(0), Y(1)) = (0, \mathbf{1}\{X = 1\}),$$

and then drawing $A \mid X = k \sim \mathsf{Bernoulli}(\pi_k)$*.*

Proposition 1. [prop:two-category-confounding] (Two-Category Confounding Witness) *For* ϵ*, suppose:*

- ***(Real parameter.)*** $\epsilon \in \mathbb{R}$*.*
- ***(Positive contrast.)*** $0 < \epsilon$*.*
- ***(Overlap range.)*** $\epsilon \le 1/2$*.*
- ***(Witness law.)*** *Q is the full-data law constructed in definition 15, and P is its observed marginal law.*

Then Q satisfies assumption 3 and assumption 4, the observed law P satisfies assumption 2 with overlap constant ϵ, and the ATE functional of definition 6 obeys

$$\tau(P) = \frac{1}{2}, \qquad E_P[Y \mid A = 1] - E_P[Y \mid A = 0] = \epsilon.$$

proposition 1 separates the causal estimand from the unadjusted observed contrast. The construction satisfies the usual causal restrictions, but the treated and control groups place different weights on the two covariate categories. As a result, the naive contrast can be as small as the overlap constant even though the ATE equals one half.

3 Main results

The setup in definition 14 reduces the statistical question to a worst-case mean-squared-error problem over finite-alphabet overlap classes. The main theorem identifies the rate in the fixed-interior case $0 < \epsilon < 1/2$, for large enough n and within the calibrated growth range $d \leq \rho_\epsilon n \log n$. For each fixed $0 < \epsilon < 1/2$ under the calibrated range, the nonparametric term is $d^2/(n^2 \log^2 n)$, reflecting the cost of binary outcomes and unrestricted covariate heterogeneity in this finite-alphabet benchmark. The lower-bound scale is imported from the fixed-sample construction of Zeng et al. [2024]; this paper proves the matching computable hybrid upper bound.

Theorem 1. [`thm:sharp-minimax-fixed-interior`] (Fixed-Interior Minimax Rate)[1] *Fix $0 < \epsilon < 1/2$. Then there exist constants $a_\epsilon, \rho_\epsilon, C_\epsilon > 0$ and an integer $N_\epsilon < \infty$ such that the following hold.*

- ***(Sharp sandwich.)*** *For every n, d, every single-observation law P, and every sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ as in definition 3, if $d > 0$, $n \geq N_\epsilon$, and*

$$d \leq \rho_\epsilon n \log n,$$

then, with $\mathsf{R}_{n,d,\epsilon}$ as in definition 14 and $\widehat{\tau}_n^{\mathrm{hyb}}$ as in definition 11,

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\} \leq \mathsf{R}_{n,d,\epsilon} \leq \sup_{Q^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_Q \left[\left(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(Q) \right)^2 \right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

- ***(Parametric window.)*** *For every $K > 0$, there exists $C_K > 0$ such that, for every n, d, every single-observation law P, and every sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, if $d > 0$, $n \geq N_\epsilon$,*

$$d \leq K \sqrt{n} \log n \qquad \text{and} \qquad d \leq \rho_\epsilon n \log n,$$

then

$$\frac{a_\epsilon}{n} \leq \mathsf{R}_{n,d,\epsilon} \leq \frac{C_K}{n}.$$

- ***(Consistency threshold.)*** *For any sequences n_j, d_j with $n_j \to \infty$, eventually $d_j > 0$, and eventually*

$$d_j \leq \rho_\epsilon n_j \log n_j,$$

one has

$$\mathsf{R}_{n_j,d_j,\epsilon} \to 0 \quad \Longleftrightarrow \quad \frac{d_j}{n_j \log n_j} \to 0.$$

theorem 1 gives three consequences under the displayed fixed-interior range. First, throughout $d \leq \rho_\epsilon n \log n$, the minimax risk is of exact order

$$n^{-1} + \frac{d^2}{n^2 \log^2 n}.$$

[1] **Formalization scope.** This result uses the published conclusion [Zeng et al., 2024] (Theorem 2; Appendix C.5; Lemma 4; Appendix D.2). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

Second, the parametric rate remains attainable when $d = O(\sqrt{n}\log n)$. Third, within the same displayed range, consistent estimation is possible exactly when $d = o(n\log n)$. Thus the logarithmic factor changes the consistency boundary as well as the constants. The constants in the theorem may depend on the fixed overlap margin, while the estimator in definition 11 uses the same formula across ϵ.

The next theorem records the hybrid computability statement, the fixed-interior upper envelope, and the separate behavior near the randomized endpoint. The hybrid part uses the same estimator for all fixed-interior overlap classes. For each fixed $0 < \epsilon < 1/2$, under $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, the selected rule in definition 13 is only an oracle-calibrated deterministic comparison between that fixed-interior guarantee and the centered linear estimator from definition 12. Polynomial approximation and adaptation ideas of this form are standard in nonsmooth functional estimation [Cai and Low, 2011, Lepski et al., 1999, Jiao et al., 2015], but here they are applied to the four-cell ATE contribution in definition 5.

Theorem 2. [`thm:overlap-adaptive-universal-hybrid`] (Universal Hybrid Envelope) *The following conclusions hold for the hybrid estimator $\widehat{\tau}_n^{\mathrm{hyb}}$ of definition 11 and the centered estimator $\widehat{\tau}_{\mathrm{ctr}}$ of definition 12, with minimax risk $\mathsf{R}_{n,d,\epsilon}$ as in definition 14.*

- ***(Computability.)*** *There is a constant $K \in \mathbb{N}$, $K > 0$, such that for every $n, d \in \mathbb{N}$ there is a real-arithmetic program which, on the hybrid count input, returns exactly $\widehat{\tau}_n^{\mathrm{hyb}}$ for every sample $O_1, \dots, O_n$, and whose arithmetic/comparison operation count is at most $K\, d\, M(n)^4$.*

- ***(Fixed-overlap hybrid calibration and envelope.)*** *For every $0 < \epsilon < 1/2$, there exist constants $C_\epsilon > 0$, $\rho_\epsilon > 0$, and $N_\epsilon \in \mathbb{N}$ such that, for all $n, d \in \mathbb{N}$ with $n > 0$, $d > 0$, $n \geq N_\epsilon$, and*

$$d \leq \rho_\epsilon n \log n,$$

the hybrid estimator satisfies

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(P))^2\right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

For the deterministic selected estimator obtained from the rule that chooses $\widehat{\tau}_n^{\mathrm{hyb}}$ when

$$C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\} \leq \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and otherwise chooses $\widehat{\tau}_{\mathrm{ctr}}$, one also has

$$\mathsf{R}_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2\right]$$

and

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2\right] \leq \max\{C_\epsilon, 4\} \left\{ \frac{1}{n} + \min\left[\frac{d^2}{n^2(\log n)^2}, (1/2-\epsilon)^2 \right] \right\}.$$

- ***(Centered estimator.)*** *For every $0 < \epsilon \leq 1/2$ and every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$,*

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_P\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right] \leq \frac{1}{n} + 4(1/2 - \epsilon)^2.$$

- ***(Endpoint bracket.)*** *For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$,*

$$\frac{1}{100n} \leq \mathsf{R}_{n,d,1/2} \leq \frac{1}{n}.$$

theorem 2 separates three issues that are often conflated. The hybrid estimator is computable from the finite set of split counts, with operation count polynomial in $M(n)$ and linear in the alphabet size. For each fixed $0 < \epsilon < 1/2$, the same hybrid formula attains the fixed-overlap upper rate in theorem 1. Near $\epsilon = 1/2$, however, the centered estimator has the sharper envelope $n^{-1} + 4(1/2 - \epsilon)^2$, and at the endpoint $\epsilon = 1/2$ the minimax risk is bracketed at the parametric order n^{-1}. The deterministic selected rule uses the displayed theoretical constants and is therefore a comparison device, not an adaptive estimator for unknown overlap or unknown calibration constants.

4 Discussion and extensions

The fixed-interior result extends the usual semiparametric picture. For fixed d, or more generally throughout the parametric window in theorem 1, the minimax risk is of order n^{-1}, in line with the classical efficiency theory for the average treatment effect under overlap [Hahn, 1998, Hirano et al., 2003]. Allowing the alphabet to grow freely adds the challenge of estimating many small, poorly sampled conditional contrasts, reflected by the additional term $d^2/(n^2(\log n)^2)$.

This distinction explains the hybrid estimator's improvement over direct plug-in and weighting estimators in the unrestricted discrete model. A plug-in estimator based on empirical conditional means treats each category as if its arm-specific denominators were reliable. In light categories, those denominators are often zero or very small, and trimming or stabilization introduces bias whose worst-case size depends on the number of categories. Inverse-probability weighting faces the same local denominator problem from the other side: bounded population overlap in assumption 2 coexists with unstable empirical treatment-cell counts when category mass is small. The hybrid estimator in definition 11 uses ratios on pilot-certified heavy categories and a polynomial continuation on the light categories. The sharp logarithmic factor in theorem 1 confirms that this distinction is essential to the rate.

Doubly robust and targeted estimators are designed to protect first-order inference against nuisance estimation error, and they are central in modern observational causal inference [Robins et al., 1994, Bang and Robins, 2005, van der Laan and Rubin, 2006]. Their usual guarantees are most naturally expressed through convergence conditions on nuisance regressions or propensities. The warning of Robins and Ritov [1997] that unrestricted high-dimensional adjustment can obstruct uniform model-free inference is therefore especially relevant here. The experiment class in definition 3 imposes no smoothness, sparsity, margin, or ordering condition that would make such nuisance rates available uniformly. The minimax statement isolates that difficulty in a finite-alphabet benchmark: for each fixed $0 < \epsilon < 1/2$ under the calibrated range, binary outcomes and unrestricted covariate heterogeneity yield the term $d^2/(n^2(\log n)^2)$.

Higher-order influence-function methods are also closely related in motivation, since they seek to reduce bias from estimating high-dimensional nuisance components [Robins et al., 2008, 2009, 2017, Liu et al., 2017, Kennedy et al., 2024]. The present construction is complementary: it replaces unstable light-cell ratios directly by a polynomial approximation to the cell functional in definition 5. The rate in theorem 1 is achieved by the hybrid estimator, whose light-cell branch uses a degree-$M(n)$ polynomial approximation and factorial-moment plug-in terms.

This result is related to recent discussions of limited overlap and high-dimensional adjustment [D'Amour et al., 2021, Balakrishnan et al., 2023, Jin and Syrgkanis, 2025]. Theorem 1 fixes $0 < \epsilon < 1/2$ and shows that many covariate categories with small marginal mass remain difficult under bounded overlap. Theorem 2 separately characterizes the exact-randomization endpoint $\epsilon = 1/2$, where the minimax risk is bracketed between constants times n^{-1}, uniformly in d.

4.1 Limitations and future work

The selected-envelope part of theorem 2 connects these two regimes without claiming a full transition theory or an adaptive implementation. For each fixed interior $0 < \epsilon < 1/2$, and under the theorem's hybrid calibration range $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, it compares the fixed-overlap hybrid estimator with the centered estimator using the known theoretical constants and gives an upper bound involving

$$\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, (1/2-\epsilon)^2\right\}.$$

This is useful as a theoretical envelope when the assignment mechanism is close to randomized, because the centered estimator improves as ϵ approaches 1/2. The selector depends on ϵ and C_ϵ, so it should not be read as a data-driven procedure when those quantities are unknown. Moreover, the theorem does not assert that this combined upper envelope is minimax sharp for triangular arrays with $\epsilon = \epsilon_n$. The fixed-interior lower bound applies for each fixed $0 < \epsilon < 1/2$, and the endpoint bracket applies at $\epsilon = 1/2$; together they leave open the possibility that the exact minimax transition depends on the joint rates at which d, n, and $1/2 - \epsilon_n$ vary.

Appendices

A proofs and auxiliary lemmas

This appendix records the auxiliary statements used to assemble the upper and lower risk bounds in theorem 1 and theorem 2. The organizing distinction is the one built into definition 11: pilot-certified heavy categories are handled by empirical ratios, while pilot-certified light categories are handled by polynomial continuation and factorial moments. The statements below are given as mathematical inputs to the main theorems, with external approximation and minimax ingredients cited where they enter.

The light-cell part is where the logarithmic factor enters the upper bound. Polynomial approximation of nonsmooth distribution functionals has long been used to improve over direct plug-in estimators in large-alphabet problems [Paninski, 2003, Valiant and Valiant, 2011, 2017, Orlitsky et al., 2016, Jiao et al., 2015, Wu and Yang, 2016, Han et al., 2015, 2020, Wu and Yang, 2019]. Here the same principle is applied to the homogeneous four-cell contribution in definition 5. The factorial-moment representation in definition 11 turns the polynomial $P_{M,B}$ into an estimable light-category contribution without dividing by the small empirical treatment counts that make naive ratios unstable.

Lemma 1. [`lem:light-cell-polynomial`] (Light-Cell Polynomial Rate) *Fix an overlap constant ϵ with $0 < \epsilon \leq 1/2$. The hybrid estimator is computable in the real-arithmetic model: there* *is a universal operation-count constant $K > 0$ such that, for every n and d, a real-arithmetic program reading the hybrid count vector returns $\widehat{\tau}_n^{\mathrm{hyb}}$ exactly and uses at most $KdM(n)^4$ arithmetic and comparison operations.*

Moreover, there exist constants $C_\epsilon, c_\epsilon > 0$, depending only on ϵ, such that for every sample size n, alphabet size d, law P on $\{1, \ldots, d\} \times \{0,1\} \times \{0,1\}$, and sample law $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ as in definition 3, if

$$d \leq c_\epsilon n \log n,$$

then the light-cell contribution satisfies

$$E_{P^{\otimes n}}\left[\left\{\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)\right\}^2\right] \leq C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Proof of lemma 1. The computability assertion is obtained by expanding the hybrid rule of definition 11 into a real-arithmetic expression tree, including the threshold comparisons and the final truncation. The resulting program reads the hybrid count vector, evaluates exactly $\widehat{\tau}_n^{\mathrm{hyb}}$, and has operation count bounded by $450\, d\, M(n)^4$. Thus the first assertion holds with the universal constant $K = 450$.

For the risk bound, take

$$K_c = 4(e+1)8192^2 + 16 \cdot 8192^2, \qquad K_b = \left(\frac{65536}{\epsilon \alpha_0^2}\right)^2, \qquad K_f = 8(e+1)8192^2$$

and $C_\epsilon = 3(K_c + K_b + K_f)$, while $c_\epsilon = 1$. These constants are positive and depend only on ϵ. After replacing the sample law by $P^{\otimes n}$ using definition 3, split into the two cases $n \geq N_0$ and $n < N_0$. If $n < N_0$, the light set is empty by construction, so both $\widehat{T}_{\mathcal{L}}$ and its target sum vanish and the bound is immediate from nonnegativity of the rate.

Assume now $n \geq N_0$. The light error decomposes pointwise as

$$\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) = X + Y + Z,$$

where X is the centered fixed-light factorial-polynomial term on genuinely light cells, Y is the selected genuine-light approximation bias, and Z is the selected false-light error. The centered term has second moment at most $K_c\{n^{-1} + d^2/(n^2 \log^2 n)\}$, after transporting the finite product integral to the infinite iid product space. The false-light term is bounded similarly by $K_f\{n^{-1} + d^2/(n^2 \log^2 n)\}$. For the bias term, the definition of the genuine-light set gives each genuinely light cell mass at most $B/4$, hence its cell vector has total mass at most B. The Chebyshev approximation bound gives the per-cell error $2B/(\epsilon M^2)$; the light-selection indicator is bounded by one, and there are at most d cells. The calibration inequalities then convert the squared summed bias into

$$Y^2 \leq K_b \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}$$

pointwise, and integrating preserves the same bound. Finally,

$$(X + Y + Z)^2 \leq 3(X^2 + Y^2 + Z^2),$$

so adding the three estimates gives the asserted inequality with C_ϵ. ∎

lemma 1 has two roles. Its computability clause gives an explicit finite algorithm whose operation count is polynomial in the approximation degree after the split counts are formed. Its risk clause controls the stochastic error and approximation bias of the light side at the target scale. The degree $M(n) \asymp \log n$ is what produces the denominator $(\log n)^2$ in the squared nonparametric term.

The pilot split must classify categories well enough that the two estimators are used on their intended mass ranges. The next statement gives the required sandwich: pilot-heavy categories have population mass above a lower threshold, while pilot-light categories have population mass below an upper threshold, except on an event whose probability is polynomially small.

Lemma 2. [`lem:pilot-sandwich`] (Pilot Sandwich Bound) *Let $L = \log(en)$. For a sample satisfying assumption 1 with single-observation law P, let $\widehat{\mathcal{H}}$ be the pilot heavy set at threshold t, let $\widehat{\mathcal{L}} = \widehat{\mathcal{H}}^c$, and write $m_0 = |I_0|$, $B_- = tL/(2m_0)$, and $B_+ = 2tL/m_0$.*

For every $K > 0$ and $c > 0$, there exist constants $t_0 > 0$, $C > 0$, and an integer cutoff N such that the following holds. If

- ***(Threshold.)*** *$t \geq t_0$;*
- ***(Sample size.)*** *$n \geq N$;*
- ***(Dimension.)*** *d is a nonnegative integer;*
- ***(Law.)*** *P is a law on the finite observation alphabet $\{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$;*
- ***(Growth.)*** *$d \leq c\, n\, L$;*
- ***(Sampling.)*** *the sample law satisfies assumption 1 with single-observation law P,*

then

$$\Pr\Big(\widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq B_-\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq B_+\}\Big) \leq Cn^{-K}.$$

Moreover, for every $c > 0$, there exist $C > 0$ and an integer cutoff N such that the following holds with $t = 256$. If

- ***(Sample size.)*** *$n \geq N$;*
- ***(Dimension.)*** *d is a nonnegative integer;*
- ***(Law.)*** *P is a law on the finite observation alphabet $\{1, \dots, d\} \times \{0, 1\} \times \{0, 1\}$;*
- ***(Growth.)*** *$d \leq c\, n\, L$;*
- ***(Sampling.)*** *the sample law satisfies assumption 1 with single-observation law P,*

then

$$\Pr\Big(\widehat{\mathcal{H}} \not\subseteq \{k : p_k \geq 256L/(2m_0)\} \text{ or } \widehat{\mathcal{L}} \not\subseteq \{k : p_k \leq 512L/m_0\}\Big) \leq Cn^{-4}.$$

Proof of lemma 2. For a fixed category k, the pilot count on I_0 is exactly a Bernoulli count with success probability p_k and $m_0 = |I_0|$ trials. If the sandwich event fails, then for some k either $p_k < tL/(2m_0)$ while the pilot count exceeds tL, or $p_k > 2tL/m_0$ while the pilot count is at most tL. This is just the definition of the pilot heavy set together with the floor inequalities. The first alternative is bounded by the Bernoulli upper-tail estimate

$$\exp\{-tL(\log 2 - 1/2)\},$$

and the second by the lower-tail estimate

$$\exp\{-tL/4\}.$$

A union bound over the d categories therefore gives, for $n \geq 2$ and $t \geq 0$,

$$\Pr(\text{pilot bad}) \leq d\left[\exp\{-tL(\log 2 - 1/2)\} + \exp\{-tL/4\}\right].$$

Since $\log 2 - 1/2 > 0$, choose

$$t_0 = \max\left\{\frac{K+4}{\log 2 - 1/2}, 4(K+4)\right\}, \qquad C = 2c, \qquad N = 2.$$

For every $t \geq t_0$, both exponential factors are at most $\exp\{-(K+4)L\}$. Using $d \leq cnL$, $L \leq n$, and $\exp\{-q\log(en)\} = e^{-q}n^{-q}$, the preceding display is bounded by $2c\,n^{-K}$. Replacing the sample law by $P^{\otimes n}$ under iid sampling proves the first assertion.

For the stated numerical specialization, take $K = 4$ and $t = 256$. The inequalities $8 \leq 256(\log 2 - 1/2)$ and $8 \leq 256/4$ satisfy the same threshold requirements, so the preceding argument gives constants $C = 2c$ and $N = 2$, after harmless enlargement, and hence the displayed n^{-4} bound. ☐

The constants in lemma 2 are deliberately one-sided. The estimator uses the stated separation to justify ratio bounds on the heavy side and polynomial bounds on the light side. The special threshold $t = 256$ is the calibration used in definition 11.

Once the pilot split supplies that separation, the heavy categories can be treated by ordinary ratio estimation. Overlap controls the arm-specific denominators in population, and the pilot event keeps the selected category masses large enough for the second-split empirical ratios to have the required mean-squared error.

Lemma 3. [`lem:universal-heavy-cell-rate`] (Heavy Cell Rate) *For every $0 < \epsilon < 1/2$, there exist constants $C_\epsilon, \rho_\epsilon > 0$ and an integer $N_\epsilon < \infty$ such that the following holds. For every $n, d \in \mathbb{N}$, every single-observation law P on $\{1, \ldots, d\} \times \{0,1\} \times \{0,1\}$, and every sample law μ_n, assume:*

- ***(Sample size.)*** *$n \geq N_\epsilon$.*
- ***(Dimension growth.)*** *$d \leq \rho_\epsilon n \log n$.*
- ***(Experiment class.)*** *$\mu_n = P^{\otimes n}$ belongs to $\mathcal{E}_{n,d,\epsilon}$ in the sense of definition 3.*

Then

$$\int \left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k)\right)^2 d\mu_n \leq C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Proof of lemma 3. Apply the $t = 256$, $K = 4$ pilot bound of lemma 2 with growth constant 1. It gives constants $C_{\text{bad}} > 0$ and N_{bad} such that the pilot-failure contribution is at most $C_{\text{bad}}n^{-4}$ whenever $d \leq n\log(en)$. Set

$$K_\epsilon = \frac{16}{\epsilon} + 12 + \frac{1}{\epsilon^4}, \qquad C_\epsilon = 4K_\epsilon + C_{\text{bad}}, \qquad \rho_\epsilon = 1,$$

and take $N_\epsilon = \max\{8, N_0, N_{\text{bad}}\}$.

Fix n, d, P, μ_n satisfying the hypotheses. The product-law assumption replaces μ_n by $P^{\otimes n}$, and $d \le n \log n \le n \log(en)$ puts the pilot bound in force. On the pilot-good event, every pilot-heavy category has

$$p_k \ge \frac{256 \log(en)}{2m_0}.$$

Thus the random heavy set is eligible for the fixed-heavy-set arm estimate with $B = 256 \log(en)/(2m_0)$. For any deterministic eligible set H, the ratio arm error is bounded by a residual term and a missing-outcome term; the residual uses overlap and the missing term uses the inverse-count bound. Together with the empirical-mass variance term this gives the fixed-arm bound, and the elementary algebra of the split sizes reduces it to

$$K_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

Fiber the pilot-good event according to the realized heavy set; the fixed-set estimate applies on each fiber and sums to the same bound for each arm of the random heavy set. Since the heavy contribution is the treated arm contribution minus the control arm contribution, the squared heavy error is at most twice the treated-arm squared error plus twice the control-arm squared error, so the pilot-good integral is at most $4K_\epsilon$ times the target rate.

On the pilot-bad event, both the heavy estimator and its target have absolute value at most one, so the squared error is at most 4; hence the bad-event integral is bounded by $C_{\text{bad}} n^{-4}$, and $n^{-4} \le n^{-1} \le n^{-1} + d^2/(n^2 \log^2 n)$. Adding the integrals over the bad event and its complement gives the asserted bound with $C_\epsilon = 4K_\epsilon + C_{\text{bad}}$. □

lemma 3 matches the light-cell rate from lemma 1. Combining the two controls the untruncated sum $\widehat{T}_{\mathcal{H}} + \widehat{T}_{\mathcal{L}}$; the final truncation in definition 11 cannot increase squared error against an ATE lying in the binary-outcome range.

The lower bound is transferred from the one-sided subclass in which only treated conditional means vary. This subclass is contained in the observational ATE experiment class and identifies the same hard functional through $\tau(P)$. The external minimax input is the fixed-sample large-alphabet lower bound of Zeng et al. [2024]; standard lower-bound comparisons then pass it to definition 14, in the same spirit as classical minimax reductions [Tsybakov, 2009, Cai and Low, 2011].

Lemma 4. [`lem:ate-lower-bound-transfer`] (ATE Minimax Lower Bound)[2] *For any $0 < \epsilon < 1/2$, let $\mathsf{R}_{n,d,\epsilon}$ denote the minimax mean-squared-error risk of definition 14. There exist constants $a_\epsilon, b_\epsilon \in \mathbb{R}$ and a natural-number cutoff N_ϵ such that $a_\epsilon > 0$, $b_\epsilon > 0$, and for every $n, d \in \mathbb{N}$ with $d > 0$, $n \ge N_\epsilon$, and*

$$d \le b_\epsilon n \log n,$$

one has

$$\mathsf{R}_{n,d,\epsilon} \ge a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}.$$

[2]**Formalization scope.** This result uses the published conclusion [Zeng et al., 2024] (Theorem 2; Appendix C.5; Lemma 4; Appendix D.2). The cited source proof is not formalized here: Lean verifies its use and all remaining steps. If that cited conclusion is false, the portions of this result that depend on it are not certified.

Proof of lemma 4. Fix $0 < \epsilon < 1/2$. By the cited fixed-sample control-zero lower bound [Zeng et al., 2024], there are constants $a_\epsilon, b_\epsilon > 0$ and N_ϵ such that, for every n, d,

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\}$$

is no larger than the treated-arm minimax risk over the subclass with $\mu_{0k} = 0$ whenever $0 < d$, $N_\epsilon \le n$, and $d \le b_\epsilon n \log n$.

On that control-zero subclass, definition 6 reduces $\tau(P)$ law by law to the treated-arm functional, since the control regression mean is zero in every category. It remains to compare the treated-arm minimax risk on this subclass with the unrestricted ATE minimax risk. For any fixed measurable estimator, every control-zero law is also a law in $\mathcal{E}_{n,d,\epsilon}$, and the preceding target identity makes its treated-arm loss equal to its ATE loss. Thus the subclass worst-case risk is bounded by the unrestricted worst-case risk.

Passing from fixed estimators to the estimator infimum gives that the treated-arm minimax risk over the control-zero subclass is no larger than $\mathsf{R}_{n,d,\epsilon}$. Combining this comparison with the cited lower bound gives, for the same constants $a_\epsilon, b_\epsilon, N_\epsilon$,

$$a_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2 (\log n)^2} \right\} \le \mathsf{R}_{n,d,\epsilon}$$

for all n, d satisfying $0 < d$, $N_\epsilon \le n$, and $d \le b_\epsilon n \log n$. □

lemma 4 is the only auxiliary statement in this section whose substantive lower-bound scale is imported from the published large-alphabet ATE construction [Zeng et al., 2024]. Its conclusion is stated directly for the minimax risk in definition 14, so it can be paired with the hybrid upper bound without changing the experiment class.

The remaining statements describe the randomized endpoint and the envelope near it. At exact randomization, treatment assignment carries no category-specific imbalance, and the centered linear estimator in definition 12 becomes the natural comparison estimator.

Lemma 5. [`lem:near-randomization-linear-upper`] (Centered Linear MSE Bound) *For any $n, d \in \mathbb{N}$, any $\epsilon \in \mathbb{R}$, any single-observation law P, and any sample law μ_n, suppose that*

- ***(Sample size.)*** $n > 0$.
- ***(Experiment class.)*** *μ_n belongs to the experiment class $\mathcal{E}_{n,d,\epsilon}$ generated by P, as in definition 3.*

Define

$$\widehat{\tau}_{\mathrm{ctr}} = \frac{1}{n} \sum_{i=1}^{n} 2(2A_i - 1) \left(Y_i - \frac{1}{2} \right).$$

Then, with $\tau(P)$ as in definition 6,

$$\mathbb{E}_{\mu_n} \left[\left(\widehat{\tau}_{\mathrm{ctr}} - \tau(P) \right)^2 \right] \le \frac{1}{n} + 4 \left(\frac{1}{2} - \epsilon \right)^2.$$

Proof of lemma 5. Let

$$S(O) = 2(2A - 1) \left(Y - \frac{1}{2} \right).$$

The estimator is the sample average $n^{-1}\sum_{i=1}^{n} S(O_i)$, and $S(O)^2 = 1$ for every binary observation; hence $\mathrm{Var}_P(S) \le 1$. Under the product law in definition 3, independence gives

$$\mathrm{Var}_{\mu_n}(\widehat{\tau}_{\mathrm{ctr}}) \le \frac{1}{n}.$$

It remains to bound the bias. For every positive-mass category,

$$E[S(O) \mid X = k]p_k = p_k(\mu_{1k} - \mu_{0k}) + 2(\pi_k - \tfrac{1}{2})p_k(\mu_{1k} + \mu_{0k} - 1),$$

and zero-mass categories contribute zero. Summing over k gives

$$E_P S(O) - \tau(P) = \sum_k 2(\pi_k - \tfrac{1}{2})p_k(\mu_{1k} + \mu_{0k} - 1).$$

By overlap, $|\pi_k - \frac{1}{2}| \le \frac{1}{2} - \epsilon$, while $\mu_{1k}, \mu_{0k} \in [0,1]$ implies $|\mu_{1k} + \mu_{0k} - 1| \le 1$, and $\sum_k p_k = 1$. Therefore

$$|E_P S(O) - \tau(P)| \le 2\left(\frac{1}{2} - \epsilon\right).$$

The mean of $\widehat{\tau}_{\mathrm{ctr}}$ is $E_P S(O)$, and the usual identity

$$E[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2] = \mathrm{Var}(\widehat{\tau}_{\mathrm{ctr}}) + \{E\widehat{\tau}_{\mathrm{ctr}} - \tau(P)\}^2$$

then yields

$$E_{\mu_n}\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right] \le \frac{1}{n} + 4\left(\frac{1}{2} - \epsilon\right)^2.$$

□

lemma 5 quantifies the cost of using the centered estimator away from exact randomization. The variance term is parametric, while the bias term is controlled by the distance from $\epsilon = 1/2$. This bound complements the fixed-interior hybrid analysis in overlap classes close to randomized treatment.

A matching parametric lower bound at the endpoint follows already from a one-category binary-outcome subproblem.

Lemma 6. [`lem:one-category-bernoulli-lower`] (Bernoulli Randomization Lower Bound) *For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, the minimax mean-squared-error risk $\mathsf{R}_{n,d,1/2}$ of definition 14 satisfies*

$$\mathsf{R}_{n,d,1/2} \ge \frac{1}{100n}.$$

Proof of lemma 6. Fix n, d with $n > 0$ and $d > 0$, and put

$$\delta = \frac{2/5}{\sqrt{n}}.$$

Then $\delta \ge 0$, $\delta^2 = 4/(25n)$, and $\delta \le 2/5$, since $\sqrt{n} \ge 1$. Thus both binary means $1/2$ and $1/2 + \delta$ are valid probabilities. Consider the two product laws $P_0^{\otimes n}$ and $P_1^{\otimes n}$ in which X is uniform on the d-point alphabet, $A \mid X \sim \mathsf{Bernoulli}(1/2)$, the control outcome mean is $1/2$, and the treated outcome mean is $1/2$ under P_0 and $1/2+\delta$ under P_1. Both observed laws satisfy exact overlap with $\epsilon = 1/2$, hence both product laws belong to $\mathcal{E}_{n,d,1/2}$. Their ATE values are

$$\tau(P_0) = 0, \qquad \tau(P_1) = \delta.$$

Moreover,

$$n\frac{(1/2)\delta^2}{(1/2)(1-1/2)} = \frac{8}{25} \le \log 2,$$

so the parametric total-variation comparison gives

$$\mathrm{TV}(P_0^{\otimes n}, P_1^{\otimes n}) \le \frac{1}{2}.$$

Now let $\widehat{\tau}$ be any measurable estimator. Since the two ATE values are separated by δ, the two-point testing bound with radius $\delta/2$ and total variation at most $1/2$ gives

$$\max_{j=0,1} P_j^{\otimes n}\left(|\widehat{\tau} - \tau(P_j)| \ge \frac{\delta}{2}\right) \ge \frac{1}{4}.$$

For each $j = 0, 1$,

$$\left(\frac{\delta}{2}\right)^2 P_j^{\otimes n}\left(|\widehat{\tau} - \tau(P_j)| \ge \frac{\delta}{2}\right) \le E_{P_j^{\otimes n}}\left[(\widehat{\tau} - \tau(P_j))^2\right],$$

by integrating the squared error over the event on which it is at least $(\delta/2)^2$. Hence

$$\max_{j=0,1} E_{P_j^{\otimes n}}\left[(\widehat{\tau} - \tau(P_j))^2\right] \ge \frac{\delta^2}{16} = \frac{1}{100n}.$$

Since $P_0^{\otimes n}$ and $P_1^{\otimes n}$ are members of $\mathcal{E}_{n,d,1/2}$, the worst-case risk of this arbitrary estimator is at least $1/(100n)$. Taking the infimum over all measurable estimators in definition 14 proves the claim.

□

The one-category reduction in lemma 6 removes the large-alphabet feature entirely, leaving the ordinary difficulty of estimating a Bernoulli mean contrast under randomized assignment. Combining it with the centered-estimator upper bound gives the endpoint bracket.

Lemma 7. [`lem:randomized-endpoint-minimax`] (Randomized Endpoint Minimax Risk) *For every $n, d \in \mathbb{N}$ with $n > 0$ and $d > 0$, the minimax mean-squared-error risk of definition 14 at exact randomization satisfies*

$$\frac{1}{100n} \le \mathsf{R}_{n,d,1/2} \le \frac{1}{n}.$$

Proof of lemma 7. The lower bound is exactly lemma 6.

For the upper bound, take the centered estimator $\widehat{\tau}_{\mathrm{ctr}}$ as a measurable candidate in definition 14. The infimum over all measurable estimators is therefore no larger than its worst-case risk:

$$\mathsf{R}_{n,d,1/2} \le \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,1/2}} E_{P^{\otimes n}}\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right].$$

It remains to bound this worst-case risk. If $\mathcal{E}_{n,d,1/2}$ is empty, the supremum bound is immediate. Otherwise, for each $P^{\otimes n} \in \mathcal{E}_{n,d,1/2}$, lemma 5 gives

$$E_{P^{\otimes n}}\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right] \leq \frac{1}{n} + 4\left(\frac{1}{2} - \frac{1}{2}\right)^2 = \frac{1}{n}.$$

Taking the supremum over the class yields $\mathsf{R}_{n,d,1/2} \leq 1/n$, which together with the lower bound proves the two-sided bracket.

$\square$

lemma 7 shows that the endpoint $\epsilon = 1/2$ is qualitatively different from the fixed-interior large-alphabet problem: uniformly in d, the risk remains of order n^{-1}. The next lemma combines this observation with the hybrid upper bound through the oracle-calibrated selected estimator introduced in definition 13.

Lemma 8. [`lem:combined-upper-envelope`] (Combined Upper Envelope) *Fix constants $\epsilon, C_\epsilon, \rho_\epsilon \in \mathbb{R}$ and $N_\epsilon \in \mathbb{N}$. Assume:*

- ***(Interior overlap.)*** $0 < \epsilon < 1/2$.
- ***(Positive constants.)*** $C_\epsilon > 0$ *and* $\rho_\epsilon > 0$.
- ***(Hybrid bound.)*** *For every $n, d \in \mathbb{N}$, if $0 < n$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$, then the hybrid estimator $\widehat{\tau}_n^{\mathrm{hyb}}$ of definition 11 satisfies*

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(P)\right)^2\right] \leq C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Then for every $n, d \in \mathbb{N}$ with $0 < n$, $N_\epsilon \leq n$, and $d \leq \rho_\epsilon n \log n$, let $K_\epsilon = \max\{C_\epsilon, 4\}$. Define the C_ϵ, ϵ-dependent selected estimator $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$ to be $\widehat{\tau}_n^{\mathrm{hyb}}$ when

$$C_\epsilon \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \leq \frac{1}{n} + 4(1/2 - \epsilon)^2$$

and to be $\widehat{\tau}_{\mathrm{ctr}}$ otherwise. This selected estimator satisfies

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}}\left[\left(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P)\right)^2\right] \leq K_\epsilon \left[\frac{1}{n} + \min\left\{\frac{d^2}{n^2(\log n)^2}, (1/2 - \epsilon)^2\right\}\right].$$

Proof of lemma 8. Fix n, d satisfying the displayed hypotheses, and abbreviate

$$a = \frac{1}{n}, \qquad u = \frac{d^2}{n^2(\log n)^2}, \qquad v = \left(\frac{1}{2} - \epsilon\right)^2, \qquad K_\epsilon = \max\{C_\epsilon, 4\}.$$

First, the centered estimator satisfies

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} E_{P^{\otimes n}}\left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2\right] \leq a + 4v.$$

Indeed, if the class is empty this is immediate, while otherwise it follows pointwise from lemma 5 and then by taking the supremum.

By the assumed hybrid bound, the hybrid estimator has worst-case risk at most $C_\epsilon(a+u)$. The selected estimator is the hybrid estimator exactly when $C_\epsilon(a+u) \le a+4v$, and is the centered estimator otherwise. Therefore, in the two cases respectively,

$$\sup_{\mathcal{E}_{n,d,\epsilon}} E\Big[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2\Big] \le \min\{C_\epsilon(a+u),\, a+4v\}.$$

The same selected estimator is a measurable candidate in definition 14, so the minimax risk is no larger than its worst-case risk; this is just the infimum comparison over measurable estimators.

It remains to compare the last minimum with the asserted envelope. Since $C_\epsilon \le K_\epsilon$, $4 \le K_\epsilon$, and $a, u, v \ge 0$, if $u \le v$, then

$$\min\{C_\epsilon(a+u), a+4v\} \le C_\epsilon(a+u) \le K_\epsilon(a+u) = K_\epsilon\{a+\min(u,v)\}.$$

If $v \le u$, then

$$\min\{C_\epsilon(a+u), a+4v\} \le a+4v \le K_\epsilon(a+v) = K_\epsilon\{a+\min(u,v)\}.$$

Substituting back the definitions of a, u, v gives the claimed bound for $\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon}$.

□

lemma 8 formalizes the oracle-calibrated deterministic upper-envelope comparison between the fixed-interior hybrid estimator and the centered near-randomization estimator. Consequently, theorem 2 gives a theoretical risk guarantee near $\epsilon = 1/2$. Data-driven adaptation and the sharp triangular-array minimax transition are identified as future directions in the limitations section.

B verification note

This appendix documents the scope and boundary of the machine-checked component of the paper.

The checked development includes the finite-alphabet observed-data model used in assumption 1 and definition 3. In particular, the single-observation law P, the iid product law $P^{\otimes n}$, and the finite four-cell representation by category are represented directly. The same layer also includes the bridge from finite products to the infinite-product notation used for sequences of observations, so that the sample-level risk statements are connected to repeated iid sampling rather than to an informal limiting construction.

The local observational model is checked at the level needed for the overlap and ATE objects. The overlap cone in definition 4, the homogeneous cell functional in definition 5, and the ATE functional in definition 6 are represented as finite-alphabet objects with arbitrary category masses. Thus the verification covers the unrestricted discrete-confounding geometry used throughout the paper, including categories with small marginal probability, subject only to the stated overlap restrictions.

The estimator objects are also included in the checked scope. This covers the deterministic split, the split cell counts, the heavy and light pilot sets, the ratio contribution on pilot-certified heavy categories, the Chebyshev-polynomial continuation on light categories, the factorial-moment estimator, and the final truncation in definition 11. The minimax risk in definition 14 is checked as a worst-case mean-squared-error criterion over the experiment class, with the infimum taken over measurable estimators as stated there.

Lean declaration names and low-level convention checks are retained in the formal crosswalk and verification artifacts. The statistical proof prose focuses on the heavy/light decomposition, approximation bias, variance control, minimax reduction, and endpoint comparison.

For results whose theorem-local verification footnotes cite an external mathematical input, the cited input is part of the paper's stated dependency boundary. The machine check covers the formal statement that imports that input, the compatibility of its hypotheses with the objects defined in the paper, and the remaining derivation from that point. Results without such a footnote are checked internally against the definitions and assumptions stated in the paper.

All theorem statements and formal derivations in this paper are machine-checked in Lean 4, subject to the theorem-local external inputs explicitly identified in the formalization-scope footnotes.

C Proofs of the main results

Proof of theorem 1.

1. First record the upper estimate for the hybrid estimator. Lemma 1 gives constants $C_L, c_L > 0$ for the light-cell error, and lemma 3 gives constants $C_H, c_H > 0$ and a cutoff N_U for the heavy-cell error. Set

$$C_U = 2(C_L + C_H), \qquad \rho_U = \min\{c_L, c_H\}.$$

For any admissible product law satisfying $n \geq N_U$ and

$$d \leq \rho_U n \log n,$$

the light and heavy bounds apply simultaneously. Pointwise, the truncation in $\widehat{\tau}_n^{\mathrm{hyb}}$ can only reduce squared error because $\tau(P) \in [-1, 1]$. Also

$$\sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k) + \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) = \tau(P),$$

and hence

$$(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(P))^2 \leq 2\left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k)\right)^2 + 2\left(\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k)\right)^2.$$

After integration and applying the two component bounds,

$$E_P\left[(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(P))^2\right] \leq C_U \left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Taking the supremum over the experiment class gives the same bound for the worst-case risk of $\widehat{\tau}_n^{\mathrm{hyb}}$, and the minimax risk is no larger because the hybrid estimator is one admissible estimator.

2. The lower-bound input is lemma 4, obtained by transferring the fixed-sample one-arm lower bound of [Zeng et al., 2024, Theorem 2; Appendix C.5; Lemma 4; Appendix D.2]. It gives constants $a > 0$, $\rho_L > 0$, and $N_L < \infty$ such that, whenever $d > 0$, $n \geq N_L$, and

$$d \leq \rho_L n \log n,$$

one has

$$\mathsf{R}_{n,d,\epsilon} \geq a\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

Now define

$$a_\epsilon = a, \qquad \rho_\epsilon = \min\{\rho_U, \rho_L\}, \qquad C_\epsilon = C_U, \qquad N_\epsilon = \max\{N_U, N_L\}.$$

These constants are positive, with $N_\epsilon < \infty$.

3. Under the hypotheses of the sharp sandwich, $n \geq N_\epsilon$ implies both $n \geq N_U$ and $n \geq N_L$, while

$$d \leq \rho_\epsilon n \log n$$

implies both required growth bounds, with ρ_U and with ρ_L. The lower estimate from Step 2 and the upper estimate from Step 1 therefore give

$$a_\epsilon\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \leq \mathsf{R}_{n,d,\epsilon} \leq \sup_{Q^{\otimes n}\in\mathcal{E}_{n,d,\epsilon}} E_Q\left[(\widehat{\tau}_n^{\mathrm{hyb}} - \tau(Q))^2\right] \leq C_\epsilon\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\}.$$

This proves the sharp sandwich.

4. Fix $K > 0$ and put

$$C_K = C_U(1 + K^2).$$

For the parametric window, keep the interior growth gate as part of the hypotheses: assume $d > 0$, $n \geq N_\epsilon$,

$$d \leq K\sqrt{n}\,\log n, \qquad d \leq \rho_\epsilon n \log n.$$

The first of these inequalities, together with $d > 0$ and $K > 0$, forces $n > 0$ and $\log n > 0$. Squaring the $K\sqrt{n}\log n$ bound and dividing by the positive denominator gives

$$\frac{d^2}{n^2(\log n)^2} \leq \frac{K^2}{n}.$$

The lower bound uses the retained $\rho_\epsilon n \log n$ gate to invoke the lower estimate:

$$\frac{a_\epsilon}{n} \leq a_\epsilon\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \leq \mathsf{R}_{n,d,\epsilon}.$$

The upper bound uses the same gate to invoke the hybrid upper estimate, and then the displayed ratio bound:

$$\mathsf{R}_{n,d,\epsilon} \leq C_U\left\{\frac{1}{n} + \frac{d^2}{n^2(\log n)^2}\right\} \leq \frac{C_U(1+K^2)}{n} = \frac{C_K}{n}.$$

This proves the parametric window under the stated interior growth condition.

5. Finally consider sequences n_j, d_j with $n_j \to \infty$, eventually $d_j > 0$, and eventually

$$d_j \leq \rho_\epsilon n_j \log n_j.$$

Set

$$q_j = \frac{d_j}{n_j \log n_j}.$$

Since $n_j \to \infty$, we have $1/n_j \to 0$, and eventually $n_j \geq 2$, so $q_j \geq 0$. Algebraically, for all sufficiently large j,

$$\frac{1}{n_j} + \frac{d_j^2}{n_j^2(\log n_j)^2} = \frac{1}{n_j} + q_j^2.$$

For those same indices, using a canonical exact-randomization law in the class to instantiate the upper estimate, Steps 1 and 2 give the two-sided bound

$$a_\epsilon \left(\frac{1}{n_j} + q_j^2 \right) \leq \mathsf{R}_{n_j,d_j,\epsilon} \leq C_U \left(\frac{1}{n_j} + q_j^2 \right).$$

If $\mathsf{R}_{n_j,d_j,\epsilon} \to 0$, the lower bound and $a_\epsilon > 0$ imply

$$\frac{1}{n_j} + q_j^2 \to 0.$$

Then $0 \leq q_j^2 \leq n_j^{-1} + q_j^2$ eventually, so $q_j^2 \to 0$. Because $q_j \geq 0$ eventually, continuity of the square-root map gives $q_j \to 0$.

Conversely, if $q_j \to 0$, then $q_j^2 \to 0$, and together with $1/n_j \to 0$ this yields

$$\frac{1}{n_j} + q_j^2 \to 0.$$

The minimax risk is nonnegative, and the upper bound above squeezes $\mathsf{R}_{n_j,d_j,\epsilon}$ to 0. This proves the consistency threshold and completes the proof.

□

Proof of theorem 2.

1. The computability assertion is obtained by taking the universal program constant $K = 450$. For each pair (n,d), the corresponding real-arithmetic program reads the hybrid count vector, evaluates exactly $\widehat{\tau}_n^{\text{hyb}}$ on every sample $O_1, \ldots, O_n$, and has operation count at most $450\, d\, M(n)^4$. This is the required first clause, with $K = 450$.

2. Fix $0 < \epsilon < 1/2$. Lemma 1 gives constants $C_L, c_L > 0$ for the light-cell contribution, and lemma 3 gives constants $C_H, c_H > 0$ and $N_H \in \mathbb{N}$ for the heavy-cell contribution. Set

$$C_\epsilon = 2(C_L + C_H), \qquad \rho_\epsilon = \min\{c_L, c_H\}, \qquad N_\epsilon = N_H.$$

Then $C_\epsilon > 0$ and $\rho_\epsilon > 0$. If $n \geq N_\epsilon$ and $d \leq \rho_\epsilon n \log n$, then the same growth condition holds with both c_L and c_H. For every $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$,

$$\mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}_n^{\text{hyb}} - \tau(P))^2 \right] \leq 2\, \mathbb{E}_{P^{\otimes n}} \left[\left(\widehat{T}_{\mathcal{H}} - \sum_{k \in \widehat{\mathcal{H}}_n} \phi(q_k) \right)^2 \right] + 2\, \mathbb{E}_{P^{\otimes n}} \left[\left(\widehat{T}_{\mathcal{L}} - \sum_{k \in \widehat{\mathcal{L}}_n} \phi(q_k) \right)^2 \right].$$

Applying the heavy- and light-cell bounds yields

$$\mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}_n^{\text{hyb}} - \tau(P))^2 \right] \leq C_\epsilon \left\{ \frac{1}{n} + \frac{d^2}{n^2(\log n)^2} \right\}.$$

Taking the supremum over $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$ gives the stated hybrid envelope.

3. With this hybrid bound in hand, apply lemma 8. The selected estimator is an oracle-calibrated measurable deterministic competitor in the minimax infimum, hence

$$\mathsf{R}_{n,d,\epsilon} \leq \sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2 \right].$$

   Writing

$$u = \frac{d^2}{n^2(\log n)^2}, \qquad v = (1/2-\epsilon)^2, \qquad K_\epsilon = \max\{C_\epsilon, 4\},$$

   the selector first gives the risk bound $\min\{C_\epsilon(1/n+u),\, 1/n+4v\}$. Splitting into the two cases $u \leq v$ and $v < u$, and using $C_\epsilon \leq K_\epsilon$ and $4 \leq K_\epsilon$, gives

$$\sup_{P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}} \mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}^{\mathrm{sel}}_{C_\epsilon,\epsilon} - \tau(P))^2 \right] \leq \max\{C_\epsilon, 4\} \left\{ \frac{1}{n} + \min\left[\frac{d^2}{n^2(\log n)^2}, (1/2-\epsilon)^2 \right] \right\}.$$

4. Now fix $0 < \epsilon \leq 1/2$ and $n, d > 0$. If $\mathcal{E}_{n,d,\epsilon}$ is empty, the supremum bound is trivial. Otherwise, for each $P^{\otimes n} \in \mathcal{E}_{n,d,\epsilon}$, lemma 5 gives

$$\mathbb{E}_{P^{\otimes n}} \left[(\widehat{\tau}_{\mathrm{ctr}} - \tau(P))^2 \right] \leq \frac{1}{n} + 4(1/2-\epsilon)^2.$$

   Taking the supremum over the class proves the centered-estimator clause.

5. Finally, for every $n, d > 0$, lemma 7 gives directly

$$\frac{1}{100n} \leq \mathsf{R}_{n,d,1/2} \leq \frac{1}{n}.$$

   Together with the preceding clauses, this proves the theorem.

□

References

Sivaraman Balakrishnan, Edward H. Kennedy, and Larry Wasserman. The fundamental limits of structure-agnostic functional estimation. *arXiv preprint arXiv:2305.04116*, 2023. doi: 10.48550/arXiv.2305.04116. To appear in Statistical Science.

Heejung Bang and James M. Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005. doi: 10.1111/j.1541-0420.2005.00377.x.

Alexandre Belloni, Victor Chernozhukov, and Christian Hansen. High-dimensional methods and inference on structural and treatment effects. *Journal of Economic Perspectives*, 28(2):29–50, 2014. doi: 10.1257/jep.28.2.29.

T. Tony Cai and Mark G. Low. Testing composite hypotheses, hermite polynomials and optimal estimation of a nonsmooth functional. *The Annals of Statistics*, 39(2):1012–1041, 2011. doi: 10.1214/10-AOS849.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097.

Alexander D'Amour, Peng Ding, Avi Feller, Lihua Lei, and Jasjeet Sekhon. Overlap in observational studies with high-dimensional covariates. *Journal of Econometrics*, 221(2):644–654, 2021. doi: 10.1016/j.jeconom.2019.10.014.

Jinyong Hahn. On the role of the propensity score in efficient semiparametric estimation of average treatment effects. *Econometrica*, 66(2):315–331, 1998. doi: 10.2307/2998560.

Yanjun Han, Jiantao Jiao, and Tsachy Weissman. Minimax estimation of discrete distributions under <inline-formula> <tex-math notation="latex">ℓ_1 </tex-math></inline-formula> loss. *IEEE Transactions on Information Theory*, 61(11): 6343–6354, 2015. doi: 10.1109/TIT.2015.2478816.

Yanjun Han, Jiantao Jiao, and Tsachy Weissman. Minimax estimation of divergences between discrete distributions. *IEEE Journal on Selected Areas in Information Theory*, 1(3):814–823, 2020. doi: 10.1109/JSAIT.2020.3041036.

Keisuke Hirano, Guido W. Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 71(4):1161–1189, 2003. doi: 10.1111/1468-0262.00442.

Jiantao Jiao, Kartik Venkat, Yanjun Han, and Tsachy Weissman. Minimax estimation of functionals of discrete distributions. *IEEE Transactions on Information Theory*, 61(5):2835–2885, 2015. doi: 10.1109/TIT.2015.2412945.

Jikai Jin and Vasilis Syrgkanis. Structure-agnostic optimality of doubly robust learning for treatment effect estimation. *arXiv preprint arXiv:2402.14264*, 2025. doi: 10.48550/arXiv.2402.14264. To appear in COLT 2025.

Edward H. Kennedy, Sivaraman Balakrishnan, James M. Robins, and Larry Wasserman. Minimax rates for heterogeneous causal effect estimation. *The Annals of Statistics*, 52(2):793–816, 2024. doi: 10.1214/24-AOS2369.

O. Lepski, A. Nemirovski, and V. Spokoiny. On estimation of the l r norm of a regression function. *Probability Theory and Related Fields*, 113(2):221–253, 1999. doi: 10.1007/s004409970006.

Lin Liu, Rajarshi Mukherjee, Whitney K. Newey, and James M. Robins. Semiparametric efficient empirical higher order influence function estimators. *arXiv preprint arXiv:1705.07577*, 2017. doi: 10.48550/arXiv.1705.07577.

Alon Orlitsky, Ananda Theertha Suresh, and Yihong Wu. Optimal prediction of the number of unseen species. *Proceedings of the National Academy of Sciences*, 113(47):13283–13288, 2016. doi: 10.1073/pnas.1607774113.

Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6): 1191–1253, 2003. doi: 10.1162/089976603321780272.

James M. Robins and Ya'acov Ritov. Toward a curse of dimensionality appropriate (CODA) asymptotic theory for semi-parametric models. *Statistics in Medicine*, 16(3):285–319, 1997. doi: 10.1002/(SICI)1097-0258(19970215)16:3⟨285::AID-SIM535⟩3.0.CO;2-\#.

James M. Robins, Andrea Rotnitzky, and Lue Ping Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866, 1994. doi: 10.1080/01621459.1994.10476818.

James M. Robins, Lingling Li, Eric J. Tchetgen Tchetgen, and Aad W. van der Vaart. Higher order influence functions and minimax estimation of nonlinear functionals. In *Probability and Statistics: Essays in Honor of David A. Freedman*, volume 2 of *Institute of Mathematical Statistics Collections*, pages 335–421. Institute of Mathematical Statistics, Beachwood, OH, 2008. doi: 10.1214/193940307000000527.

James M. Robins, Eric J. Tchetgen Tchetgen, Lingling Li, and Aad W. van der Vaart. Semiparametric minimax rates. *Electronic Journal of Statistics*, 3:1305–1321, 2009. doi: 10.1214/09-EJS479.

James M. W. Robins, Lingling Li, Rajarshi Mukherjee, Eric J. Tchetgen Tchetgen, and Aad van der Vaart. Minimax estimation of a functional on a structured high-dimensional model. *The Annals of Statistics*, 45(5):1951–1987, 2017. doi: 10.1214/16-AOS1515.

Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41.

Donald B. Rubin. Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701, 1974. doi: 10.1037/h0037350.

Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009. doi: 10.1007/b13794.

Gregory Valiant and Paul Valiant. Estimating the unseen. In *Proceedings of the 43rd Annual ACM Symposium on Theory of Computing*, pages 685–694, 2011. doi: 10.1145/1993636.1993727.

Gregory Valiant and Paul Valiant. Estimating the unseen. *Journal of the ACM*, 64(6):37:1–37:41, 2017. doi: 10.1145/3125643.

Mark J. van der Laan and Daniel Rubin. Targeted maximum likelihood learning. *The International Journal of Biostatistics*, 2(1), 2006. doi: 10.2202/1557-4679.1043.

Yihong Wu and Pengkun Yang. Minimax rates of entropy estimation on large alphabets via best polynomial approximation. *IEEE Transactions on Information Theory*, 62(6):3702–3720, 2016. doi: 10.1109/TIT.2016.2548468.

Yihong Wu and Pengkun Yang. Chebyshev polynomials, moment matching, and optimal estimation of the unseen. *The Annals of Statistics*, 47(2):857–883, 2019. doi: 10.1214/17-AOS1665.

Zhenghao Zeng, Sivaraman Balakrishnan, Yanjun Han, and Edward H. Kennedy. Causal inference with high-dimensional discrete covariates. *arXiv preprint arXiv:2405.00118*, 2024. doi: 10.48550/arXiv.2405.00118. Version 3.